

In-class assignment # 3

PHY-905-003

Computational Astrophysics and Astrostatistics

Spring 2026

Instructions: We're going to use the functions you wrote for the Bisection and Newton's method, as well as some of the additional root finding methods in the [SciPy optimize](#) package (specifically, the methods for scalar functions).

Your goal is straightforward: try to come up with multiple functions $f(x)$ that have one or more roots in an interval $[a, b]$ and which break one or more of the following methods: the Bisection Method, Newton's Method, the Secant Method (which is Newton's method with a numerically-computed derivative – use the function you wrote last time!), and [Brent's Method](#). In this case, “break” simply means to force the method to not find an existing root in that interval within a finite number of iterations.

Experiment with several functions, intervals $[a, b]$, and all of the methods described above. What are the characteristics of functions or chosen intervals $[a, b]$ that tend to break root-finding algorithms? Are some algorithms more robust than others to choice of function and/or interval? Are any of these algorithms effectively unbreakable? Make some notes in `ANSWERS.md`, and we'll also discuss it in class.

What to turn in: Turn in `ANSWERS.md`, any source code you wrote, and any plots you created (and the scripts you used to create them). **Do not** turn in object files or executables!